

CHE 330 Week 3 Homework

Assigned 9/10/2025, Due 9/16/2025 11:59 PM

1. Compare and contrast 3 methods of polymerization.

Bulk Polymerization- Pure monomer and initiator used without solvent. Polymerization is carried out in liquid monomer.

Solution Polymerization- Monomer and initiator are dissolved in a solvent. This method has easier heat control and low viscosity, but solvent must be properly removed.

Emulsion Polymerization- Monomer and surfactant droplets are evenly dispersed in water. High conversion rate and high molecular weight

2. Highlight key differences between step and chain polymerization.

In Step growth polymerization any two molecules can react with each other. The monomer is consumed early on, molecular weight build-up is slow, reaction time is very long and there is no true termination. In contrast, only the active chain end reacts with monomers in chain growth polymerization. The monomer concentration decreases steadily, molecular weight build-up is fast, MW is not dependent on time, and requires initiation or termination,

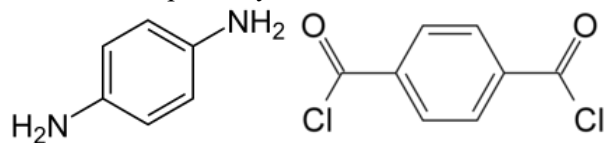
3. Identify the differences in the polymer structure, starting materials and method of formation of **Nylon 6** and **Nylon 6,6** that are produced commercially.

In the production of Nylon 6, the starting material is caprolactam and the reaction mechanism is called ring opening polymerization.

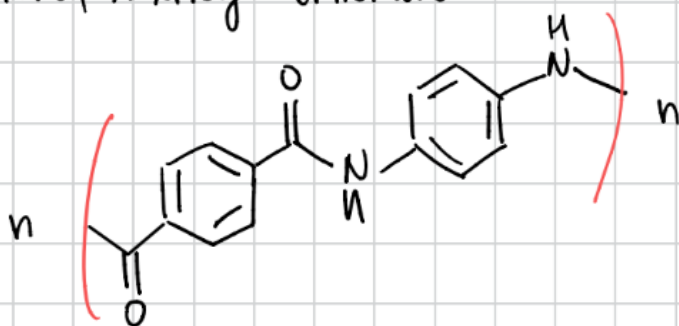
In the production of Nylon 6,6 the starting materials are adipic acid the polymer structure is described as, and the reaction mechanism is called step-growth polymerization.

The polymer structure for Nylon 6 is $[-\text{NH}-(\text{CH}_2)_5-\text{CO}-]_n$, for Nylon 6,6 it is $[-\text{NH}-(\text{CH}_2)_6-\text{NH}-\text{CO}-(\text{CH}_2)_4-\text{CO}-]_n$

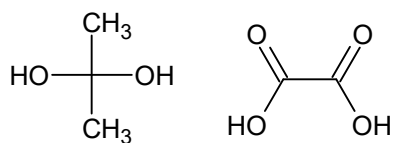
4. For each of the following monomer/starting materials, draw the polymer structure formed
p-Phenylenediamine and Terephthaloyl chloride



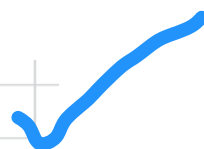
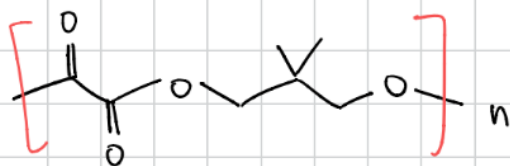
p-Phenylenediamine &
Terephthaloyl chloride



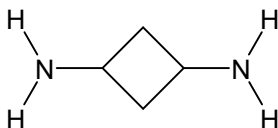
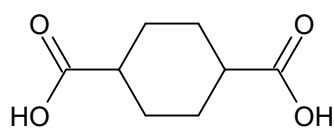
Propane-2,2-diol and oxalic acid



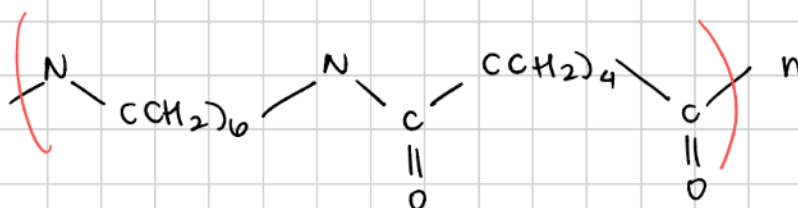
Propane-2,2 diol & oxalic acid



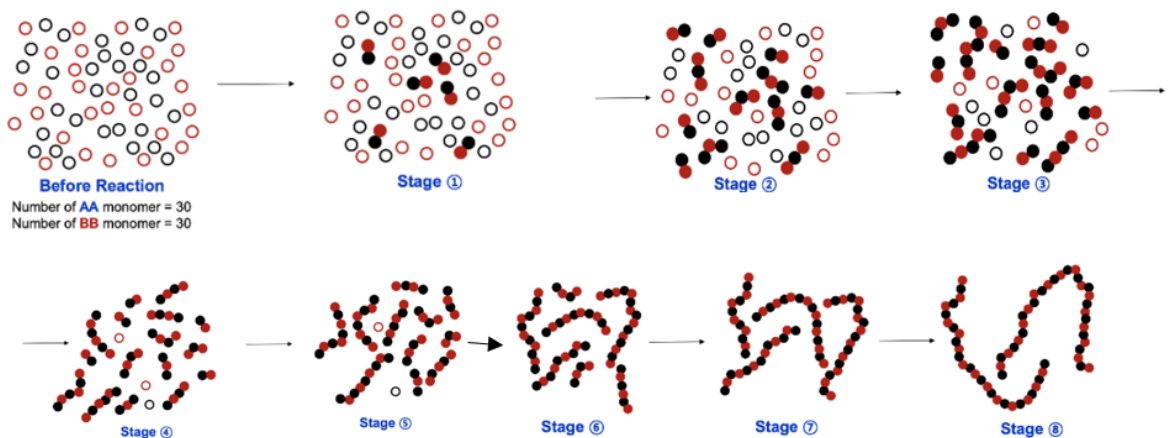
cyclohexane-1,4-dicarboxylic acid and cyclobutane-1,3-diamine



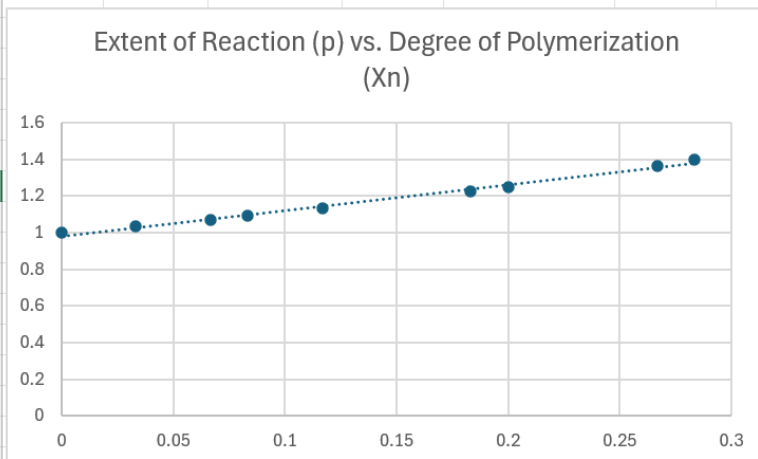
cyclohexane-1,4-dicarboxylic acid & cyclobutane-1,3-diamine



5. Consider the step polymerization of 60 bifunctional molecules shown below.
- Compute the number of functional groups at each stage of the reaction.
 - Determine the extent of the reaction and degree of polymerization at each stage of the reaction.
 - Make a plot of the extent of reaction, p (on the x-axis) vs. degree of polymerization, x_n (on the y-axis), and discuss the meaning of the plot.



	A	B	C	D	E	F	G	H
1	Stage	Reactions	Functional Groups	Extent of Reaction	Degree of Polymerization			
2	0	0	120	0	1			
3	1	5	110	0.083333	1.090909091			
4	2	12	96	0.2	1.25			
5	3	17	86	0.283333	1.395348837			
6	4	16	88	0.266667	1.363636364			
7	5	11	98	0.183333	1.224489796			
8	6	7	106	0.116667	1.132075472			
9	7	4	112	0.066667	1.071428571			
10	8	2	116	0.033333	1.034482759			



Calculations done of excel for ease. From our graph we can determine that there is positive correlation between extent of reaction and degrees of polymerization. The degree of polymerization increases gradually as the extent of reaction increases. We can see that only polymers with high molecular weight will start to form as p approaches 1.

6. Using arrow diagrams, clearly explain and illustrate the mechanism of step polymerization for a polyamide – (Nylon 6,10).

Na_2CO_3 is chosen as the base for the reaction. The nitrogen in the amine group attacks the carbon in the carbonyl group. The hydroxyl group then attracts a hydrogen attached to the nitrogen. The negative charge in the oxygen forms a double bond, the chlorine atom is then kicked off and we made our final product.

